MedSkillFlow: Liberating Medical Annotation through Autonomous Multi-Agent Orchestration

Anonymous submission

Abstract

Data quality is a primary determinant of clinical usefulness for medical vision-language models (VLMs), yet transforming heterogeneous medical corpora into effective supervision remains bottlenecked by expert labor, domain-specific constraints, and format alignment—resisting standardization and blocking rapid data–model iteration. To address these challenges, we propose MedSkillFlow, a taxonomy-guided multi-agent methodology that casts this as a structured coordination problem rather than ad hoc preprocessing. A Clinical Reasoning Taxonomy, grounded in clinical competency requirements, serves as the control interface for planner-mediated routing over reusable cognitive agents covering reformulation, normalization, adaptation, and report construction. To support both controllability and scale, MedSkillFlow couples fine-grained skill execution with reusable execution policies synthesized from taxonomy-conditioned skill chains, and uses competency-level feedback from downstream tasks to identify and refine weak supervision regions. The resulting methodology improves the adaptability of medical vision-language models across clinical tasks and supervision formats while reducing dependence on manual expert intervention. Empirically, it enables Qwen3-8B-VL to surpass the prior medical VLM (Lingshu-7B) by at least 5.6 points and outperform all proprietary baselines at 32B scale.

Introduction

Medical vision-language models (VLMs) have made rapid progress (Li et al. 2023; Zhang et al. 2024a; Lin et al. 2025), yet downstream performance remains heavily constrained by the quality of task-specific fine-tuning data. In the era of large-scale pre-trained models, data curation should no longer be regarded as an engineering prerequisite external to modeling, but rather as an integral component of the scientific problem that these models aim to solve. In medical settings, useful supervision must be clinically valid, supported by image evidence, stylistically aligned with the target task protocol, and matched to the reasoning competencies that downstream evaluation actually measures.

In practice, none of the existing approaches fully satisfy these requirements (Irvin et al. 2019; Li et al. 2023; Cui et al. 2024; Lau et al. 2018): supervision construction remains external to the learning problem. Medical corpora span heterogeneous formats—free-text radiology reports (Johnson et al. 2019), semi-structured annotations (Irvin et al. 2019), diagnostic summaries (Cai et al. 2025; Zhang et al. 2024b)—whose divergent schemas preclude direct use as supervision (Zhang et al. 2024a). Manual curation demands radiologist involvement that does not scale (Irvin et al. 2019); single-prompt LLM generation conflates formatting objectives and drifts from clinical conventions (Li et al. 2023; Zhang et al. 2024b); independent per-sample transformations repeatedly reconstruct the same task logic and can yield inconsistent supervision across corpora (Cui et al. 2024); and rule-based converters are cheap but rigid (Lau et al. 2018). What unites these approaches is that data curation operates independently of the clinical competencies the model must acquire.

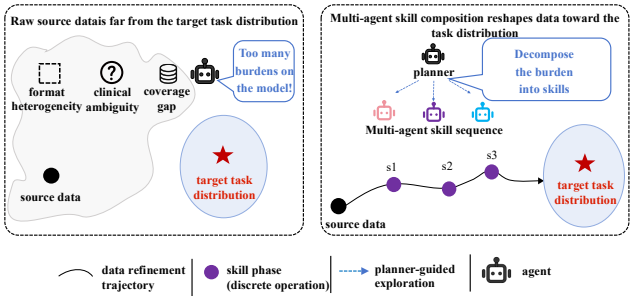


Figure 1: Motivations for MedSkillFlow. A planner composes specialized skills to resolve format, clinical, and coverage mismatches and progressively reshape source data toward the target-task distribution.

When this burden falls entirely on the model, it must simultaneously resolve format heterogeneity, clinical ambiguity, and competency coverage—a demand that scales poorly with task complexity. We argue it should be redistributed to the data stage. By encoding clinical reasoning requirements as structural priors for data synthesis, we decompose competency-level challenges traditionally deferred to model optimization into discrete cognitive operations, allowing specialized agents to address them collaboratively before training. Recent advances in LLM-based agent systems provide

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the architectural foundation for this redistribution (Yao et al. 2023; Wu et al. 2023; Shinn et al. 2023).

This perspective motivates **MedSkillFlow**, a multi-agent framework that introduces three mutually reinforcing ideas: (i) a Clinical Reasoning Taxonomy $\mathcal{T}$ that formalizes query conventions, answer norms, and competency-level priors as an explicit specification for supervision construction; (ii) a skill orchestration architecture that factorizes processing into reusable cognitive skills, assembled by a planner and dispatched to specialized agents under sample-wise or reusable policy execution; and (iii) a competency-aware self-improvement loop driven by aggregate downstream task feedback. With 3.0M taxonomy-processed samples across six benchmarks, the resulting models advance beyond prior medical VLMs at 8B scale and surpass all tested proprietary models at 32B.

Related Work

Recent medical VLMs improve how models learn from predefined supervision: VILA-M3 integrates domain experts, MIMO adds pixel grounding, LLaDA-MedV adopts diffusion modeling, and MedQwen routes low-rank experts (Nath et al. 2025; Chen et al. 2025; Dong et al. 2026; Nejatimanzari et al. 2026; Wang et al. 2026; Guo et al. 2025; Zou and Yin 2025); CheXmix, MediVLM, and GEMeX further specialize radiology pretraining, report generation, and explainable VQA (Kumar et al. 2026; Goswami, Subedi, and Chakraborty 2025; Liu et al. 2025a; Lee et al. 2024). These advances change model training or inference, but not how heterogeneous corpora are converted into consistent, competency-controlled supervision. Data-centric methods generate instructions or VQA pairs in LLaVA-Med and PMC-VQA, scale mixtures in MedMax and GMAI-VL-5.5M, evolve prompts in MMEvol, or synthesize pretraining data (Li et al. 2023; Zhang et al. 2024b; Bansal et al. 2025; Li et al. 2026; Luo et al. 2025b; Liu et al. 2025b; Sun et al. 2024a; Zhou et al. 2024; Pandita, Keniston, and Madhuripan 2025); BioMed-VITAL and Dr-LLaVA add clinical preferences or symbolic grounding (Cui et al. 2024; Sun et al. 2024b). Yet their supervision remains organized around individual datasets, outputs, or fixed tasks, limiting how directly it supports model-side transfer across task families. MedSkillFlow addresses this upstream supervision-design problem with one Clinical Reasoning Taxonomy that controls format, competency coverage, and distributional alignment across corpora, so the model can focus on learning clinically grounded reasoning patterns instead of adapting to inconsistent schema and format noise.

ReAct and AutoGen establish general planning and coordination mechanisms (Yao et al. 2023; Wu et al. 2023). In medicine, ReasonMed generates reasoning traces, MMedAgent routes tools, MedRAX reasons over chest X-rays, and MedVR and MMedAgent-RL study agentic visual reasoning and collaboration (Sun et al. 2025; Li et al. 2024; Fallahpour et al. 2025; Jiang et al. 2026; Xia et al. 2026; Wang et al. 2025; Xia et al. 2025). EyecareGPT and MAGEN instead construct specialty-focused corpora (Li et al. 2025b,c). These systems mainly target inference, textual traces, or individual specialties. MedSkillFlow composes taxonomy-conditioned skills for heterogeneous multimodal corpora, routes them to competency-specialized agents, and executes the same skill specification sample-wise or as a reusable policy.

Reflexion, LLM-as-a-judge, and MMedPO refine trajectories, judge outputs, or align factual preferences (Shinn et al. 2023; Li et al. 2025a; Zhu et al. 2025c), while A3-Tune and MedVP steer visual attention or generation (Chang et al. 2025; Zhu et al. 2025b; Luo et al. 2025a; Liu et al. 2026). These mechanisms operate on individual trajectories or outputs. MedSkillFlow instead aggregates downstream task performance at the competency level: systematic weaknesses trigger targeted augmentation, while Task Distribution Alignment constrains corpus shift.

Method

Framework Overview

As shown in Figure 2, MedSkillFlow comprises three modules: a *Raw Data Module* with Clinical Reasoning Taxonomy, a *Multi-Agent Core* with Orchestrator, skill library, and execution agents, and a *Self-Improvement Module* that feeds competency-level diagnostic signals back for iterative refinement.

Let $\mathcal{D}_s = \{(x_i, m_i)\}_{i=1}^{N}$ denote a source corpus of N samples, where x_i is the textual annotation (e.g., a diagnostic report or free-text label) and m_i is the associated medical image. Let $\mathcal{T}$ denote a target Clinical Reasoning Taxonomy (defined below). The system constructs a processed training set $\mathcal{D}_t = \{(q_j, a_j, m_j, c_j)\}_{j=1}^{M}$, where q_j is the generated question, a_j the corresponding answer, m_j the source image, and $c_j \in \mathcal{C}$ the competency dimension (e.g., modality recognition, diagnostic reasoning). We formulate data processing as optimizing a policy Π over the processing workflow (skill selection, agent routing, execution mode), balancing competency coverage against computational cost under clinically-aware quality constraints:

$$\Pi^* = \arg\max_{\Pi} \lambda_1 \mathrm{Coverage}(\Pi, \mathcal{T}) + \lambda_2 \mathrm{Align}(\Pi, \mathcal{T}) + \lambda_3 \mathrm{Validity}(\Pi) + \lambda_4 \mathrm{Scale}(\Pi),$$

where $\mathrm{Coverage}$ measures how completely Π spans the competency categories in $\mathcal{T}$, Align quantifies distributional proximity between generated and target data, $\mathrm{Validity}$ scores clinical and formatting correctness, Scale penalizes excessive computational cost, and λ_1–λ_4 are weighting coefficients.

Clinical Reasoning Taxonomy

To align supervision construction with the clinical reasoning capabilities required by downstream tasks, we encode each target task as an operational specification $\mathcal{T} = (\mathcal{Q}, \mathcal{A}, \mathcal{C}, \mathcal{P})$. $\mathcal{Q}$ defines admissible query families and their structural constraints, including open-ended, closed, multiple-choice, and report-oriented requests and, when applicable, option cardinality. $\mathcal{A}$ defines the corresponding output schemas and normalization requirements, such as answer type, canonical terminology, response granularity, and required report sections. $\mathcal{C}$ provides the competency labels assigned to generated samples, including anatomical localization, pathology identification, and modality interpretation. Finally, $\mathcal{P} = \{P_d^{\mathcal{T}}\}_{d \in \mathcal{E}}$

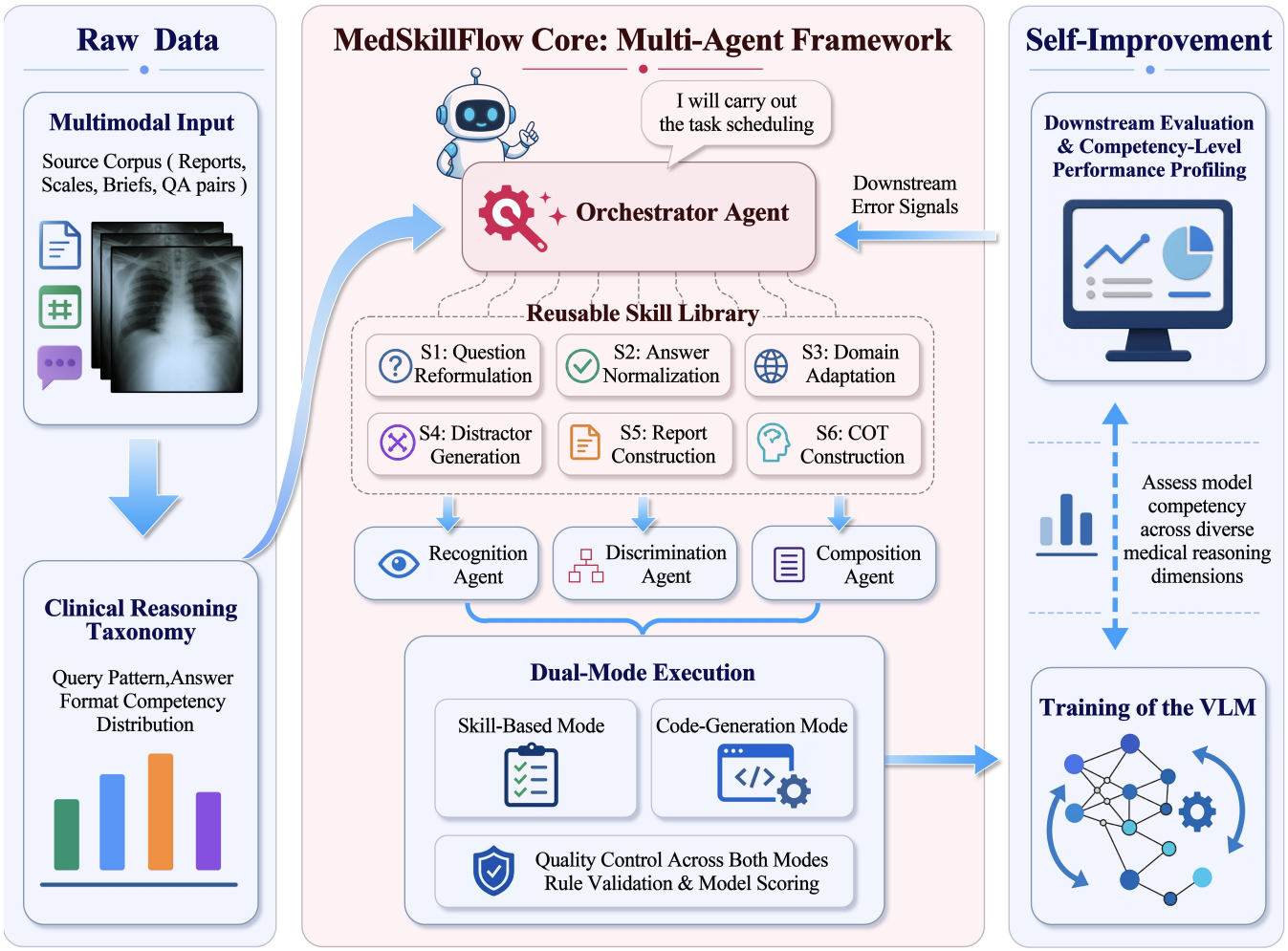


Figure 2: Overview of MedSkillFlow. The framework comprises three modules: a Raw Data Module that abstracts source corpora through a Clinical Reasoning Taxonomy $\mathcal{T}$, a Multi-Agent Core where an Orchestrator Agent coordinates a reusable skill library $\mathcal{S}$ and task-specific execution agents under dual-mode processing, and a Self-Improvement Module that profiles downstream task performance and feeds aggregate competency-level signals ϵ back to the Orchestrator.

specifies target marginal profiles over observable task dimensions $\mathcal{E}$, such as competency, query type, answer form, and clinical intent. These components are directly consumed by the pipeline: $\mathcal{Q}$ and $\mathcal{A}$ instantiate skill preconditions, output constraints, and deterministic quality-control rules; $\mathcal{C}$ conditions agent routing and competency tagging; and $\mathcal{P}$ guides sampling and supplies the target profiles used for distribution monitoring. Thus, $\mathcal{T}$ acts as a declarative contract for constructing well-formed supervision rather than only a conceptual categorization.

Multi-Agent Architecture

The Orchestrator Agent serves as the central planner that translates the taxonomy $\mathcal{T}$ into executable processing decisions, ensuring that each sample is transformed under the same clinical reasoning priors that define the target supervision. Given a source sample (x_i, m_i), the taxonomy $\mathcal{T}$, and a competency diagnostic signal ϵ from the self-improvement loop described below, it applies a routing function:

$$f_{\text{orch}}(\mathcal{T}, x_i, m_i, \epsilon) \;\rightarrow\; (\pi_i,\ \alpha_i),$$

where π_i is the ordered skill chain selected for sample i and $\alpha_i \in \{\text{Recog}, \text{Discrim}, \text{Compos}\}$ designates the execution agent responsible for carrying out π_i.

The skill library $\mathcal{S} = \{s_1, \ldots, s_6\}$ provides six reusable cognitive operations: s_1 Question Reformulation, s_2 Answer Normalization, s_3 Domain Adaptation, s_4 Distractor Generation, s_5 Report Construction, and s_6 Chain-of-Thought Augmentation. Each skill is formally a transformation $s_l : (x, m, \mathcal{T}) \rightarrow x'$ that takes a source annotation x, its paired image m, and the taxonomy $\mathcal{T}$ as input and produces a processed output x'. For each sample, the Orchestrator instantiates $\pi_i = \pi(\mathcal{T}, x_i, m_i, \epsilon) = [s_{i_1}, \ldots, s_{i_k}]$ with $k \leq |\mathcal{S}|$, mitigating *attention collapse across objectives* by distributing processing across specialized skills executed in sequence rather than requiring a single-pass generator to

satisfy all constraints simultaneously.

Once assembled, the skill chain is routed to one of three execution agents, each targeting a core scientific capability of medical imaging models. The **Recognition Agent** handles the extraction and verbalization of clinically salient findings from visual input (typically $[s_1, s_2, s_3]$). The **Discrimination Agent** operates under constrained decision settings where the model must differentiate among competing diagnostic hypotheses (typically $[s_1, s_2, s_4]$). The **Composition Agent** synthesizes image-supported observations into structured clinical narratives (typically $[s_3, s_5]$).

Dual-Mode Skill Execution and Quality Control

Skill chains are executed in one of two modes. In **sample-wise skill execution**, each sample is processed through its assigned chain, retaining intermediate outputs and fine-grained control over every operation. In **reusable policy execution**, samples sharing the same taxonomy specification and source schema are grouped by their selected skill chain. The Orchestrator then compiles the shared skill specification and representative source samples into an executable transformation policy. The policy therefore preserves the semantics and ordering of the assigned skills rather than being authored as an independent script, and applies the same validated transformation logic consistently across the corresponding corpus subset.

Conceptually, reusable policy execution is related to executable-program reasoning and code-as-policy approaches (Chen et al. 2023; Liang et al. 2023), but its role here is corpus construction: LLM reasoning compiles the taxonomy and selected skill specifications into a reusable transformation policy, which is then executed deterministically. This makes the policy taxonomy-conditioned, skill-derived, auditable, and revisable rather than a fixed hand-written schema converter. Although reusable execution improves consistency and scale, an erroneous skill output or induced policy can propagate unsupported content across many samples. We therefore place a layered quality-control gate after both execution modes to screen source-inconsistent, malformed, or clinically implausible outputs before they enter the training corpus:

$$V_{\text{rule}}(x') = \prod_{r \in \mathcal{R}} r(x'), \qquad r(x') \in \{0, 1\},$$

$$\text{Accept}(x') \iff V_{\text{rule}}(x') = 1 \ \wedge \ V_{\text{model}}(x') \geq \tau.$$

Here $\mathcal{R}$ is the set of deterministic constraints instantiated from $\mathcal{T}$, making any rule violation a hard rejection rather than allowing it to be offset by a model score. V_{model} then aggregates plausibility judgments from multiple LLMs to identify semantic and clinical inconsistencies that explicit rules cannot cover, and τ is its acceptance threshold. As an external complement to this layered control, clinicians manually reviewed a sampled subset of the retained data and found the questions and answers clinically plausible and consistent with their source medical content.

Self-Improvement and Distribution Alignment

A single processing round may not uniformly cover all competency dimensions in $\mathcal{T}$, as source corpora are inherently imbalanced and initial skill chains may under-represent certain reasoning patterns. We therefore introduce a closed-loop, competency-aware refinement mechanism that iteratively diagnoses and repairs these gaps. After each construction round, the resulting model is profiled on downstream tasks, and per-competency performance $\rho(c)$ is estimated by aggregating task-level scores over samples assigned to category c. These aggregate signals identify weak competencies as:

$$\mathcal{W} = \{c \in \mathcal{C} \mid \rho(c) < \rho_{\min}\},$$

where $\rho_{\min}$ is a pre-defined performance floor. $\mathcal{W}$ is fed back as ϵ to the Orchestrator, triggering targeted augmentation that biases skill selection toward under-performing categories. The training set is updated iteratively via

$$\mathcal{D}_t^{(r+1)} = \mathcal{D}_t^{(r)} \cup \Delta\mathcal{D}(\mathcal{W}),$$

where $\Delta\mathcal{D}(\mathcal{W})$ is generated by re-invoking the Orchestrator with $\epsilon = \mathcal{W}$. The loop runs until $\mathcal{W} = \emptyset$ or a maximum round count is reached. To prevent iterative augmentation from distorting the corpus distribution, we monitor Task Distribution Alignment (TDA):

$$\text{TDA}(\mathcal{D}_t, \mathcal{T}) = 1 - \textstyle\sum_d w_d \cdot D_{\text{JS}}(P_d^{\text{gen}} \| P_d^{\mathcal{T}}),$$

where d ranges over observable dimensions (competency, query type, answer form, clinical intent), w_d are importance weights, P_d^{gen} is the generated marginal, $P_d^{\mathcal{T}}$ is the corresponding target profile instantiated by $\mathcal{P}$, and D_{JS} is Jensen–Shannon divergence. TDA ≈ 1 indicates high corpus-level alignment, complementing the sample-level quality control described above.

Experiments

Experimental Setup

Data Construction. MedSkillFlow processes three large-scale medical corpora—MedMAT, PubMedVision (Chen et al. 2024), and MedTrinity-25M (Xie et al. 2025)—into approximately 3.0M taxonomy-aligned training samples under the guidance of $\mathcal{T}$, spanning competency-targeted diagnostic queries and structured clinical interpretation at a roughly 2:1 ratio. Potential overlap between the construction corpora and evaluation datasets is screened using exact QA, report, full-text, image-path, image-level duplicate, and frequent-template checks. No benchmark contamination was detected across the six evaluation datasets.

Task–Metric Alignment. The two task families emphasize different risks, so we adopt separate metrics. For medical VQA, accuracy is used because a model must satisfy constrained response formats (open, close, or multiple-choice styles) and incorrect choices directly indicate failure of clinical reasoning inferences under this taxonomy. For report generation, we report BLEU (Papineni et al. 2002), ROUGE-L (Lin 2004), and METEOR (Banerjee and Lavie 2005) for text surface quality, and RaTEScore (Zhao et al. 2024) as the clinical-fidelity indicator that tracks entity-level correctness and report-aligned expression. We treat RaTEScore as

Models	PMC-VQA	VQA-RAD	SLAKE	PATH-VQA
Proprietary Models				
GPT-4.1	55.2	65.0	72.2	55.5
GPT-4o	49.7	61.0	71.2	55.5
GPT-5.4	57.77	68.74	71.16	45.65
Claude Sonnet 4.6	54.4	67.6	70.6	54.2
Gemini-2.5-Flash	55.4	68.5	75.8	55.4
Gemini 3 Flash	62.62	69.62	74.55	52.03
General-purpose VLMs (<10B)				
Qwen2.5VL-7B	51.9	63.2	66.8	44.1
Janus-Pro-7B	50.1	49.7	55.2	35.4
InternVL2.5-8B	51.3	59.4	69.0	42.1
InternVL3-8B	53.8	65.4	72.8	48.6
General-purpose VLMs (>10B)				
Llama3.2-11B	48.1	58.8	65.8	32.9
InternVL3-14B	54.1	66.3	72.8	48.0
Qwen3.6-27B	62.50	71.62	75.07	46.93
Qwen2.5V-32B	54.5	71.8	71.2	41.9
InternVL2.5-38B	57.2	61.4	70.3	46.9
InternVL3-38B	56.6	65.4	72.7	51.0
Qwen3-Based Reference				
Qwen3 (base)	54.13	56.76	64.84	42.77
Qwen3+Raw	45.51	43.01	38.15	35.94
Medical VLMs (<10B)				
BiomedGPT	27.6	16.6	13.6	11.3
Med-R1-2B	47.4	39.0	54.5	15.3
MedVLM-R1-2B	48.8	49.2	56.3	36.0
HealthGPT-M3	55.4	56.8	70.8	55.4
BioMediX2-8B	41.8	55.7	54.1	34.6
LLaVA-Med-7B	22.7	46.6	51.9	35.2
MedGemma-4B-IT	49.2	<u>72.3</u>	78.2	48.1
HuatuoGPT-V-7B	53.1	67.6	68.1	44.8
Lingshu-7B	<u>56.3</u>	67.9	<u>83.1</u>	<u>61.9</u>
GMAI-VL-7B	56.1	57.2	59.7	39.1
MedSkillFlow (8B, ours)	**65.49**	**76.72**	**88.92**	**67.47**
Medical VLMs (>10B)				
HealthGPT-14B	56.4	65.0	66.1	56.7
HuatuoGPT-V-34B	56.6	61.4	69.5	44.4
Lingshu-32B	<u>57.9</u>	<u>76.7</u>	<u>86.7</u>	<u>65.5</u>
MedDr-40B	13.9	65.2	66.4	53.5
MedSkillFlow (32B, ours)	**67.38**	**76.94**	**89.35**	**68.25**

Table 1: Performance comparison across proprietary, general-purpose, and medical VLMs on four medical VQA benchmarks. Qwen3 (base) denotes the untouched Qwen3 model; Qwen3+Raw denotes Qwen3 trained directly on the unprocessed data mixture without taxonomy-guided processing. **Bold** and <u>underline</u> indicate the best and second-best medical VLMs within each size subgroup.

the primary headline metric for report benchmarks to better reflect clinical usability, and report lexical metrics for comparability with prior radiology-report literature.

Evaluation. We evaluate on six benchmarks spanning two complementary task families: VQA-RAD (Lau et al. 2018), SLAKE (Liu et al. 2021), PMC-VQA (Zhang et al. 2024b), and PATH-VQA (He et al. 2021) for medical visual question answering (accuracy), and IU-Xray (Demner-Fushman et al. 2016) / MIMIC-CXR (Johnson et al. 2019) for radiology report generation (BLEU (Papineni et al. 2002), ROUGE-L (Lin 2004), METEOR (Banerjee and Lavie 2005), RaTEScore (Zhao et al. 2024)). Baselines span proprietary models (GPT-4.1[1], GPT-4o[2], GPT-5.4[3], Claude Sonnet 4.6[4], Gemini-2.5-Flash[5], Gemini 3 Flash[6]), general-purpose VLMs (InternVL2.5/3 (Zhu et al. 2025a), Qwen2.5-VL (Bai et al. 2025), Qwen3.6 (Qwen Team 2026), Llama3.2[7]), and medical VLMs (MedGemma (Sellergren, Kohlberger et al. 2025), HuatuoGPT-V (Chen et al. 2024), LLaVA-Med (Li et al. 2023), HealthGPT (Lin et al. 2025), BioMediX2 (Mullappilly et al. 2025), Lingshu (Xu et al. 2025), MedVLM-R1 (Pan et al. 2025), BiomedGPT (Zhang et al. 2024a), GMAI-VL (Li et al. 2026)). We additionally include Qwen3-VL base and Qwen3-VL trained on the raw data mixture without taxonomy-guided processing, to isolate the framework's contribution. Evaluation is performed with MedEvalKit[8].

Main Results

Medical Visual Question Answering As presented in Table 1, MedSkillFlow establishes strong results across all four VQA benchmarks, consistently outperforming both general-purpose and medical-domain baselines.

(1) Consistent Gains. On Qwen3-8B-VL, MedSkillFlow achieves 88.92% (SLAKE), 76.72% (VQA-RAD), 65.49% (PMC-VQA), and 67.47% (PATH-VQA), substantially surpassing the base model and prior medical VLMs including MedGemma-4B-IT, HuatuoGPT-V-7B, and Lingshu-7B. The gains span anatomical recognition, diagnostic reasoning, biomedical knowledge, and histopathological interpretation, indicating broad competency improvement rather than task-specific overfitting. **(2) Necessity of Processing.** Training on a raw mixture of MedMAT, PubMedVision, and MedTrinity-25M *without* processing degrades performance *below* the base model on every benchmark (e.g., SLAKE: 64.84%→38.15%), confirming that data volume alone is insufficient—and potentially harmful—without taxonomy-aligned processing. **(3) Cross-Scale.** At 32B scale, MedSkillFlow surpasses Lingshu-32B and all proprietary models on all four benchmarks, suggesting that supervision quality scales effectively with model capacity.

Radiology Report Generation We further evaluate on MIMIC-CXR and IU-Xray. Figure 3 reports RaTEScore, while detailed NLG metrics are provided in Supplementary Figure S2. MedSkillFlow (8B) achieves leading NLG metrics

[1]https://openai.com/index/gpt-4-1/

[2]https://openai.com/index/gpt-4o-and-more-tools-to-chatgpt-free/

[3]https://openai.com/index/introducing-gpt-5-4/

[4]https://www.anthropic.com/news/claude-sonnet-4-6

[5]https://blog.google/products-and-platforms/products/gemini/gemini-2-5-flash-preview/

[6]https://blog.google/products-and-platforms/products/gemini/gemini-3-flash/

[7]https://about.fb.com/ltam/news/2024/09/llama-3-2-revolucionando-la-ia-y-la-vision-de-vanguardia-con-modelos-abiertos-y-personalizables/

[8]https://github.com/alibaba-damo-academy/MedEvalKit

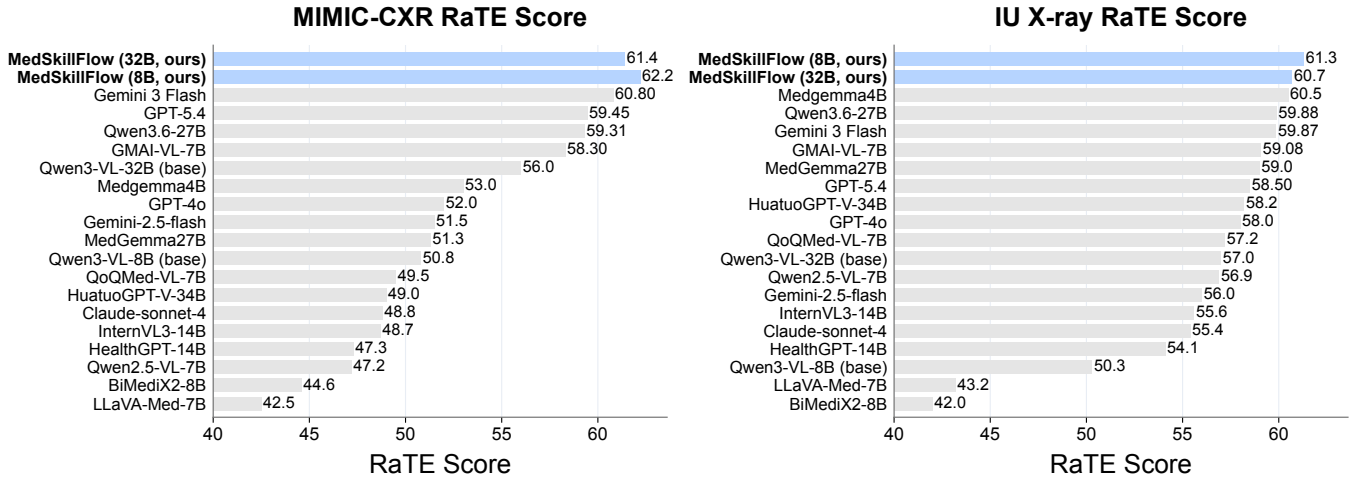


Figure 3: Clinical entity evaluation (RaTEScore) on MIMIC-CXR and IU-Xray. MedSkillFlow variants are highlighted in blue.

on both datasets. On RaTEScore, which measures clinical entity accuracy, the 8B variant reaches 62.2 on MIMIC-CXR and 61.3 on IU-Xray, outperforming all baselines shown in Figure 3 on both datasets. These results indicate that taxonomy-guided supervision improves clinical entity faithfulness alongside lexical quality.

Comparison with Hulu-Med. Hulu-Med (Jiang et al. 2025), a fully open-source medical VLM trained on 16.7M samples, is the current state of the art among $\leq$32B medical VLMs. With under one-fifth of its data (3.0M), MedSkillFlow 8B surpasses Hulu-Med-7B on SLAKE (88.92% vs. 86.8%), PATH-VQA (67.47% vs. 65.6%), and MIMIC-CXR RaTEScore (62.2 vs. 57.0), and trails by $<$1.5 points on PMC-VQA and VQA-RAD. On IU-Xray, its RaTEScore reaches 61.3 but remains below Hulu-Med-7B (67.3). Thus, with 5.6$\times$ less data, MedSkillFlow is competitive with or better than Hulu-Med on five of the six benchmarks, while performance on IU-Xray leaves room for further improvement.

Ablation Studies

We examine four pipeline dimensions on Qwen3-8B-VL in Figure 4 and separately vary the execution-agent set in Table 2.

Data Scale (a). Training on five subsets (150K–3.0M), performance generally improves with data scale (SLAKE: 81.70%→88.92%; PATH-VQA: 55.23%→67.47%), with all benchmarks remaining well above the base model even at 300K, indicating high per-sample utility and that the taxonomy imposes sufficient structural regularity to maintain quality at reduced scale. Due to training and evaluation cost, each point is evaluated with one run, but this trend is not an isolated result. It is supported by parallel evidence across 8B and 32B scales and by the sustained gains of the data-scale trajectory itself, suggesting that the observed improvements reflect the quality and quantity of taxonomy-processed supervision rather than random one-run fluctuation.

Iterative Refinement (b). Over three self-improvement rounds, SLAKE rises from 78.46% to 85.38%, VQA-RAD from 57.20% to 72.28%, and PATH-VQA from 52.29% to 60.54%. Early rounds address broadly under-covered competency regions, yielding the largest gains; later rounds target narrower gaps with diminishing but sustained returns. VQA-RAD benefits most (+15.08%), suggesting reasoning-heavy tasks are particularly responsive to competency-level augmentation.

Dataset	Raw	2-Agent	3-Agent
VQA-RAD	43.01	76.50	**76.72**
SLAKE	38.15	86.77	**88.92**
PMC-VQA	45.51	47.35	**65.49**

Table 2: Agent-number ablation on Qwen3-8B-VL (accuracy, %). Raw uses the unprocessed data mixture; the 2-Agent setting removes the Discrimination Agent; the 3-Agent setting is full MedSkillFlow. Bold indicates the best result.

Raw-Data Ablation (c). Training on unprocessed MedMAT, PubMedVision, and MedTrinity-25M (Cai et al. 2025; Chen et al. 2024; Xie et al. 2025) degrades performance below the base model on every benchmark (SLAKE: 64.84%→38.15%; VQA-RAD: 56.76%→43.01%), whereas MedSkillFlow elevates all metrics substantially, confirming that gains originate from the processing methodology rather than data volume alone.

Stage-wise Decomposition (d). Taxonomy processing yields the largest single-stage gain on knowledge-intensive benchmarks (PMC-VQA: +8.79%), self-improvement contributes most on reasoning-heavy tasks (VQA-RAD: +15.08%), and scaling to 3.0M provides a further uniform boost. The three stages exhibit complementary, non-redundant contributions, validating the multi-stage design.

Agent-Number Ablation. Table 2 isolates the contribution of the complete execution-agent set while keeping the source data fixed. The three-agent setting performs best on all three benchmarks, with the largest gain on PMC-VQA (47.35%→65.49%). This difference is consistent with PMC-

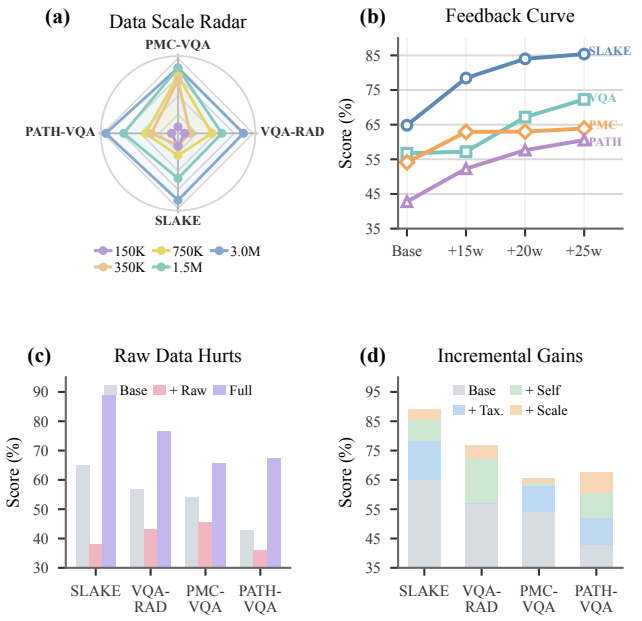


Figure 4: Ablation analysis of MedSkillFlow on four VQA benchmarks. (a) Data-scale radar over five corpus sizes from 150K to 3.0M; each spoke is independently normalized within its benchmark-specific observed range. (b) Iterative refinement: accuracy trajectories over three self-improvement rounds, with early rounds yielding the largest gains. (c) Raw-data ablation: grouped comparison of Qwen3-VL base, Qwen3-VL trained on unprocessed data, and full MedSkillFlow, revealing that unprocessed data degrades performance below the base model. (d) Stage-wise decomposition: incremental contributions of taxonomy processing, self-improvement, and data-scale expansion to the final performance.

VQA's multiple-choice format: the Discrimination Agent targets constrained decision settings and distractor-aware supervision, the capability removed from the two-agent variant. The smaller gains on VQA-RAD and SLAKE further indicate that the benefit depends on alignment between agent specialization and downstream task structure, rather than agent count alone.

Qualitative Analysis

Beyond aggregate metrics, we examine whether taxonomy-guided supervision cultivates clinical reasoning that generalizes to interactive settings. We probe the model through multi-turn interactions with progressively open-ended follow-ups demanding image-based evidence, clinical justification, and lay-language reformulation. Across scenarios, the model consistently grounds judgments in spatial anatomical relationships, articulates the clinical significance of findings rather than merely restating labels, and documents pertinent negatives in structured reports. These behaviors—image-evidence reasoning, significance-aware explanation, and systematic reporting discipline—correspond to competency dimensions encoded in $\mathcal{T}$, suggesting that taxonomy-aligned supervision instills clinical reasoning habits that transfer beyond the task formats encountered during training. Representative interactions are presented in Supplementary Figure S3.

Efficiency Analysis

We evaluate the efficiency of supervision construction under a fixed workload of 500K samples. Per-sample LLM curation requires 500K calls with approximately 800 input and 200 output tokens per call, whereas MedSkillFlow synthesizes reusable execution policies from taxonomy-conditioned skill chains in approximately 100 calls, each using about 2,000 input and 1,500 output tokens. This corresponds to approximately 500M versus 0.35M LLM tokens. Under the stated 2026 API prices, the estimated cost decreases from approximately \$2,000/\$2,700/\$120 to \$2.00/\$2.85/\$0.12 for GPT-4o, Claude Sonnet 4.6, and Gemini 2.5 Flash, respectively, representing a 947–1,000$\times$ reduction. These values are analytical estimates rather than measured billing records; Supplementary Table S1 reports the cost model, token assumptions, wall time, and human-cost accounting.

Discussion

The reported benchmarks and ablations indicate that taxonomy-conditioned supervision improves downstream performance, supporting the utility and scalability of MedSkillFlow but not the clinical acceptability of every generated sample.

The principal limitation is the absence of large-scale, systematic expert evaluation. Current human validation consists only of clinician spot-checks and therefore cannot estimate rare errors or failure rates across corpora, specialties, competency categories, and execution modes. We reduce this risk through deterministic taxonomy-derived checks, multi-LLM screening, source-consistency constraints, downstream competency profiling, and sampled clinician review. However, these safeguards cannot replace systematic expert assessment (Agrawal et al. 2025). Future work will conduct stratified, multi-reader evaluation across corpora, task families, competencies, and both execution modes, measuring clinical correctness, image and source grounding, harmful omissions, format compliance, error categories, and inter-rater agreement. Adjudicated errors will guide revisions to taxonomy rules, validation thresholds, and reusable policies. Formal de-identification remains a separate requirement.

Conclusion

In this paper, we propose MedSkillFlow, a taxonomy-guided multi-agent framework that combines a Clinical Reasoning Taxonomy, reusable cognitive skills with planner-mediated routing, sample-wise and reusable policy execution derived from the same skill chains, and competency-aware refinement guided by aggregate downstream task feedback. The framework makes medical supervision processing more controllable, scalable, and taxonomy-aligned than manual conversion or direct sample-wise prompting.

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